



## CLINICAL PROTOCOL

**Document ID: SU-014-PROT-02 | Parent Policy: SU-014 — Mechanical Thrombectomy Policy | Version: 1.0 | Controlled Copy**

Certification Alignment Note: Use the 2026 AHA/ASA AIS guideline as the controlling clinical reference. Legacy trial thresholds are reference frameworks, not automatic exclusion criteria where the 2026 guideline supports broader EVT eligibility.

# Patient Selection for Mechanical (Endovascular) Thrombectomy in Acute Ischemic Stroke

*Aligned with AHA/ASA Clinical Guidance and Stroke Center Accreditation Standards (American Stroke Association International Stroke Center Certification — Comprehensive Stroke Center & Thrombectomy-Capable Stroke Center)*

| Field               | Detail                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document type       | Institutional clinical protocol / order-set template                                                                                                                                                                                                                                                                                                                        |
| Clinical area       | Emergency Department, Neurology/Stroke Service, Neurointerventional Radiology, ICU / critical care                                                                                                                                                                                                                                                                          |
| Effective date      | _____ (institution to complete)                                                                                                                                                                                                                                                                                                                                             |
| Review cycle        | Annually, and upon publication of new AHA/ASA guideline updates                                                                                                                                                                                                                                                                                                             |
| Approved by         | Stroke Program Medical Director / Stroke Committee / Medical Executive Committee                                                                                                                                                                                                                                                                                            |
| Guideline basis     | 2026 AHA/ASA Guideline for the Early Management of Patients With Acute Ischemic Stroke (Stroke, 2026), incorporating and superseding the 2018 guideline and 2019 focused update                                                                                                                                                                                             |
| Accreditation basis | American Stroke Association International Comprehensive Stroke Center (CSC) / Thrombectomy-Capable Stroke Center (TSC) standards; American Stroke Association International Comprehensive Stroke Center Certification; AHA/ASA Get With The Guidelines–Stroke (GWTG-Stroke) registry measures; ASA/ACC/SIR/SNIS/SVIN multi-society thrombectomy device and center standards |

## 1. Purpose

To standardize the identification, imaging work-up, and selection of adult patients with acute ischemic stroke (AIS) for mechanical thrombectomy (MT), ensuring practice is concordant with current AHA/ASA guideline recommendations and with the process, timeliness, and outcome measures required for Comprehensive Stroke Center (CSC) or Thrombectomy-Capable Stroke Center (TSC) accreditation.

## 2. Scope

Applies to all patients presenting to the Emergency Department (ED), transferred from a referring facility, or developing in-hospital stroke symptoms, who are being evaluated for large-vessel-occlusion (LVO) ischemic stroke. Applies to ED, Neurology/Stroke, Neurointerventional Radiology (NIR), ICU / critical care, and stroke coordinator staff.

## 3. Accreditation Alignment Crosswalk

This protocol operationalizes the elements accrediting bodies expect a certified thrombectomy program to demonstrate:

| Accreditation domain                                         | How this protocol addresses it                                                                                                                    |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Written, guideline-based patient selection criteria          | Section 5 defines inclusion criteria mapped directly to AHA/ASA Class I/IIa recommendations, reviewed and re-approved on the cycle in Section 12. |
| 24/7/365 neurointerventional and imaging availability        | Section 6 defines activation triggers and required response-time benchmarks for CT/CTA, neurology/stroke, and the NIR team.                       |
| Standardized severe-stroke prehospital/ED notification       | Section 6 aligns with EMS large-vessel-occlusion screening tools (e.g., RACE, VAN, LAMS) and pre-notification workflow.                           |
| Timeliness (process) metrics                                 | Section 8 lists the door-to-imaging, door-to-groin-puncture, and transfer-time benchmarks tracked for GWTG-Stroke and CSC/TSC reporting.          |
| Outcome and safety metrics                                   | Section 9 lists reperfusion (mTICI), symptomatic intracranial hemorrhage (sICH), and 90-day functional outcome measures.                          |
| Multidisciplinary team and defined roles                     | Section 7 lists role-based responsibilities across ED, stroke neurology, NIR, anesthesia/sedation, and nursing.                                   |
| Case volume and credentialing                                | Section 10 references minimum annual procedure volumes and physician credentialing expectations that accrediting bodies verify at survey.         |
| Post-procedure standardized care and complication management | Section 9 covers post-thrombectomy blood pressure, imaging, and ICU-level monitoring.                                                             |
| Data submission to a national registry                       | Section 11 requires concurrent abstraction into GWTG-Stroke (or equivalent) for every thrombectomy case.                                          |
| Continuous quality improvement and periodic case review      | Section 11–12 require a standing stroke/NIR quality committee review of every case and annual protocol revision.                                  |

*Specific accreditation requirements, required metric thresholds, and survey documentation differ between American Stroke Association International Stroke Center Certification program and DNV and are updated periodically; the stroke program should verify current standards directly with the chosen certifying body before finalizing local benchmarks.*

## 4. Definitions

- LKW (Last Known Well): The most recent time the patient was witnessed at, or reported to be at, neurologic baseline.
- LVO (Large-Vessel Occlusion): Occlusion of the intracranial ICA, M1 (and select M2) segment of the MCA, basilar artery, or other segment defined as eligible below.
- ASPECTS (Alberta Stroke Program Early CT Score): 10-point NCCT scoring system for early ischemic change in the MCA territory; lower scores indicate larger established infarct core.
- mTICI: Modified Thrombolysis in Cerebral Infarction score describing angiographic reperfusion; mTICI 2b–3 defines successful reperfusion.
- mRS (modified Rankin Scale): 0–6 scale of functional disability; 0–1 = no/minimal symptoms, 0–2 = functional independence.
- Door-to-groin-puncture time: Interval from hospital arrival to first arterial (groin/radial) puncture for thrombectomy.

## 5. Patient Selection Criteria

Selection reflects the 2026 AHA/ASA acute ischemic stroke guideline, which builds on the 2018 guideline/2019 update and incorporates DAWN, DEFUSE-3, SELECT2, ANGEL-ASPECT, ATTENTION, and BAOCHE trial evidence. Local neurointerventional and stroke leadership should confirm final numeric thresholds against the current published guideline text and any subsequent living-guideline updates before adoption.

### 5.1 Universal prerequisites (all pathways)

- Age  $\geq 18$  years.
- Acute ischemic stroke with a disabling neurologic deficit attributable to a confirmed LVO on CTA/MRA (or equivalent vessel imaging).
- Intracranial hemorrhage excluded on non-contrast CT (NCCT) or MRI prior to intervention.
- Groin (or alternative arterial) access and thrombectomy can be initiated within the eligible time/imaging window.
- Treatment goals consistent with the patient's (or surrogate's) expressed wishes and code status; thrombectomy is not pursued when inconsistent with documented goals of care.

### 5.2 Early window — 0 to 6 hours from LKW (Class I)

| Criterion                   | Threshold                                                                                                                                                     |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prestroke functional status | Modified Rankin Scale 0–1 (some centers extend to selected mRS 2 patients per multidisciplinary judgment)                                                     |
| NIHSS                       | $\geq 6$ (disabling deficit)                                                                                                                                  |
| Occlusion site              | Internal carotid artery (ICA) or MCA segment 1 (M1); M2/other segments may be considered per local NIR judgment on a case-by-case, less-certain-benefit basis |
| ASPECTS                     | $\geq 6$ on NCCT (or equivalent CT perfusion/MRI core estimate)                                                                                               |
| Timing                      | Groin puncture achievable within 6 hours of LKW                                                                                                               |

### 5.3 Extended window — 6 to 16 hours from LKW (Class I)

Patients in the extended window should be selected using current 2026 AHA/ASA clinical and imaging criteria. DAWN/DEFUSE-3 mismatch criteria remain useful reference frameworks, but current eligibility is broader and should not be restricted to these legacy thresholds alone.

## 5.4 Extended window — 6 to 24 hours from LKW, DAWN-eligible (Class IIa)

Patients presenting 6–24 hours from LKW should be evaluated for thrombectomy using the current 2026 AHA/ASA criteria, including appropriate large-core and posterior-circulation pathways where applicable.

## 5.5 Extended window, large infarct core — 6 to 24 hours from LKW (Class I, 2026 update)

Based on SELECT2 and ANGEL-ASPECT, thrombectomy is now recommended for anterior-circulation LVO presenting 6–24 hours from symptom onset when ALL of the following are met:

- Age <80 years
- NIHSS  $\geq 6$
- ASPECTS 3–5 (large but not fully completed infarct core)
- No significant mass effect or midline shift on baseline imaging

*This represents a substantial expansion versus the prior (2018/2019) guideline, which generally excluded large-core strokes; confirm local imaging-interpretation criteria for “no significant mass effect” with neuroradiology.*

## 5.6 Posterior circulation — basilar artery occlusion, within 24 hours (Class I, 2026 update)

Based on ATTENTION and BAOCH, thrombectomy is recommended for patients with confirmed basilar artery occlusion who present within 24 hours of LKW/onset and have:

- NIHSS  $\geq 10$
- Mild ischemic damage on baseline imaging (e.g., posterior circulation ASPECTS or equivalent showing limited established infarct)

## 5.7 Concurrent IV thrombolysis

Eligible patients presenting within the thrombolysis window should receive IV tenecteplase (0.25 mg/kg) or alteplase (0.9 mg/kg) as first-line therapy without delaying, or waiting to assess response before proceeding to, thrombectomy in LVO patients who also meet thrombectomy criteria. Thrombolysis and thrombectomy pathways run in parallel, not sequentially.

# 6. Activation, Notification, and Imaging Workflow

## 6.1 Prehospital / ED activation triggers

- EMS use of a validated LVO screening tool (e.g., RACE, VAN, LAMS, or institution-selected equivalent) with pre-arrival notification and direct stroke-team/NIR pre-alert for positive screens.
- In systems without rapid interhospital transfer capability, direct transport to the closest thrombectomy-capable center is preferred for suspected LVO, per 2026 guideline system-of-care recommendations; where reliable rapid transfer protocols exist, transport to the closest certified stroke center with expedited transfer remains reasonable.
- Immediate “code stroke” activation on ED arrival for any patient with acute focal deficit and LKW within the eligible window.

## 6.2 Imaging pathway

- Emergent NCCT (and CT angiography head/neck,  $\pm$  CT perfusion) obtained immediately on arrival for all code-stroke activations; MRI/MRA substituted where CT perfusion is unavailable or contraindicated.
- CT perfusion or diffusion MRI mismatch imaging obtained for all patients presenting >6 hours from LKW, or with unclear/wake-up onset, to determine extended-window or large-core eligibility.
- Imaging interpreted in real time by the covering radiologist/neuroradiologist and stroke neurologist jointly with the NIR team before a treat/no-treat decision.

## 7. Multidisciplinary Roles

| Role                                                          | Responsibility                                                                                                                                                     |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ED physician/nurse                                            | Initial recognition, code stroke activation, stat imaging order, blood pressure and glucose stabilization, IV access, weight-based thrombolysis dosing if eligible |
| Stroke neurologist / vascular neurology                       | NIHSS scoring, LKW determination, eligibility review against Section 5 criteria, goals-of-care discussion, informed consent/assent for MT                          |
| Neurointerventional radiologist / neurointerventional surgeon | Final procedural eligibility confirmation, device selection, procedural execution, real-time mTICI assessment                                                      |
| Anesthesia / procedural sedation                              | Airway and hemodynamic management during thrombectomy (conscious sedation vs general anesthesia per institutional protocol and patient status)                     |
| Stroke coordinator / program manager                          | Timestamp capture, registry (GWTG-Stroke) abstraction, case log for accreditation case-volume tracking                                                             |
| ICU / critical care / post-procedure nursing                  | Post-thrombectomy neuro checks, blood pressure protocol, access-site monitoring, complication recognition                                                          |

## 8. Time-to-Treatment Benchmarks (tracked for accreditation reporting)

| Metric                                         | Target                                           | Notes                                                           |
|------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------|
| Door-to-CT completion                          | ≤25 min                                          | GWTG-Stroke / CSC benchmark                                     |
| Door-to-thrombolysis (needle) time             | ≤60 min (goal ≤45 min in ≥50% of cases)          | For thrombolysis-eligible patients                              |
| Door-to-groin-puncture (direct arrival)        | ≤90 min                                          | Institution to confirm current target against chosen accreditor |
| Door-to-groin-puncture (transfer patient)      | ≤60 min from arrival at thrombectomy center      | Reflects expedited transfer workflow                            |
| Neurointerventional team arrival/activation    | ≤30 min from activation page                     | 24/7/365 availability requirement                               |
| Recanalization (groin puncture to reperfusion) | As short as feasible; tracked, not a hard cutoff | Monitored for quality improvement                               |

*Numeric targets above reflect commonly cited stroke-system benchmarks; the stroke committee should confirm current required thresholds directly against the specific Joint Commission or DNV standard in effect at the time of adoption, as these are updated periodically.*

## 9. Post-Procedure Care and Safety Monitoring

- Admit to ICU / critical care or equivalent monitored bed after thrombectomy.
- Blood pressure: avoid intensive lowering below 140 mm Hg systolic in the first 72 hours after successful anterior-circulation LVO reperfusion, per 2026 guideline recommendations against excessive post-recanalization BP reduction; individualize for hemorrhagic transformation or other complications.
- Follow-up NCCT (typically 24 hours, or sooner if neurologic change) to assess for hemorrhagic transformation and infarct evolution.
- Standardized NIHSS reassessment at defined intervals; document any symptomatic intracranial hemorrhage (sICH).
- Access-site checks per institutional vascular access protocol.
- Dysphagia screen, VTE prophylaxis, and secondary-prevention work-up (etiology, antithrombotic planning, risk-factor management) initiated per standard post-stroke pathway.

## 10. Program, Volume, and Credentialing Requirements

Accrediting bodies verify that a program maintains defined case volumes and physician credentialing/continuing-competency standards for thrombectomy privileging. The stroke committee and medical staff office should confirm and document current minimum annual procedure volumes per physician and per program, board certification/training requirements for neurointerventionalists, and periodic peer case review, per the specific accreditor (American Stroke Association International Stroke Center Certification program CSC/TSC or DNV) and relevant specialty society (SNIS/ACR/ASNR/SVIN) standards in effect at the time of application.

## 11. Data Registry and Quality Metrics

| Metric category      | Examples tracked                                                                                             |
|----------------------|--------------------------------------------------------------------------------------------------------------|
| Process              | Door-to-CT, door-to-needle, door-to-groin-puncture, EMS pre-notification rate                                |
| Angiographic outcome | Successful reperfusion rate (mTICI 2b–3), first-pass effect rate                                             |
| Safety               | Symptomatic intracranial hemorrhage rate, procedural complication rate, access-site complications            |
| Functional outcome   | 90-day modified Rankin Scale distribution / independence rate (mRS 0–2)                                      |
| Registry submission  | Concurrent case abstraction into GWTG-Stroke (or designated equivalent registry) for every thrombectomy case |

Every thrombectomy case (treated and cases where thrombectomy was considered but not performed) should be reviewed at a standing stroke/neurointerventional quality committee meeting, with variances from this protocol documented and discussed.

## 12. Review and Revision

This protocol should be formally reviewed at least annually by the stroke program medical director and multidisciplinary stroke committee, and whenever the AHA/ASA publishes a new or living guideline update, a new pivotal trial materially changes the evidence base, or accreditation standards change. Version history and approval signatures should be maintained per institutional policy document-control requirements.

## 13. References

- Prabhakaran S, Gonzalez NR, Zachrison KS, et al. 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke: A Guideline From the American Heart Association/American Stroke Association. *Stroke*. 2026 (epub ahead of print).
- Powers WJ, Rabinstein AA, Ackerson T, et al. 2018 Guidelines for the Early Management of Patients With Acute Ischemic Stroke, and 2019 Update. *Stroke*. 2018;49:e46–e110; *Stroke*. 2019;50:e344–e418.
- Nogueira RG, et al. Thrombectomy 6 to 24 Hours after Stroke with a Mismatch between Deficit and Infarct (DAWN). *N Engl J Med*. 2018;378:11–21.
- Albers GW, et al. Thrombectomy for Stroke at 6 to 16 Hours with Selection by Perfusion Imaging (DEFUSE 3). *N Engl J Med*. 2018;378:708–718.
- Sarraj A, et al. Trial of Endovascular Thrombectomy for Large Ischemic Strokes (SELECT2). *N Engl J Med*. 2023;388:1259–1271.
- Huo X, et al. Trial of Endovascular Therapy for Acute Ischemic Stroke with Large Infarct (ANGEL-ASPECT). *N Engl J Med*. 2023;388:1272–1283.
- Tao C, et al. / Jovin TG, et al. ATTENTION and BAOCHE trials of thrombectomy for basilar artery occlusion. *N Engl J Med*. 2022.
- American Stroke Association International Stroke Center Certification program. Comprehensive Stroke Center and Thrombectomy-Capable Stroke Center Certification Requirements (current edition). DNV Healthcare USA. Comprehensive Stroke Center Certification Program (current edition).
- American Heart Association/American Stroke Association. Get With The Guidelines–Stroke Program Manual and measure specifications (current edition).
- Institutions should retrieve and cite the current, full-text versions of the above (particularly the 2026 AHA/ASA guideline and the specific accreditor's current standards manual) directly from the publisher/accreditor rather than relying on this summary alone.*